KQC7016 DATA ANALYTICS

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LAB 3 REPORT

Github Link: https://github.com/u2103136-lab/KQC7016-Data-Analytics/tree/main/Lab%203

Sri Ganesh A/L Nithiyananthan	U2103177/2
Avienash A/L Shankar	U2103136/2

Department of Electrical Engineering

Universiti Malaya

1. Introduction

Analysis of Variance (ANOVA) is a statistical technique used to determine whether significant differences exist among the means of three or more groups. Unlike multiple t-tests, which increase the risk of Type-I error, ANOVA controls the overall error rate.

In this lab, we apply ANOVA to the WorldEnergy.csv dataset to investigate whether **GDP per capita** and **time period** affect **low-carbon energy per capita** across six Southeast Asian countries which are Indonesia, Malaysia, Philippines, Singapore, Thailand, and Vietnam.

Why choose this data and analysis?

- **Dependent variable:** Low-carbon energy per capita. This reflects clean energy deployment relative to population size, capturing actual energy access and consumption patterns beyond simple percentages.
- **Independent variable 1:** GDP per capita (categorical: Low, Middle, High). Allows testing whether wealthier nations (e.g., Singapore, Malaysia) have significantly higher low-carbon energy use than lower-income neighbours (Indonesia, Philippines, Vietnam).
- **Independent variable 2:** Period (categorical: Early = 2000-2010, Recent = 2011-2024). Captures the global acceleration of renewable energy after 2010, enabling testing of whether the gap between income groups has widened or narrowed over time.
- **Interaction:** The combined effect of GDP level and Period, tested via two-way ANOVA, to see if income-based differences changed between the Early and Recent periods.
- **Justification:** Grounded in energy economics literature (positive income-clean energy relationship) and climate policy timelines (post-2010 renewable cost declines). The six Southeast Asian countries were selected for regional relevance and data completeness.

2. Code with explanation

2.1 Import Libraries and Load Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.formula.api import ols
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
```

Purpose: Imports the necessary Python libraries for data manipulation (pandas, numpy), data visualization (matplotlib, seaborn), and statistical modeling/hypothesis testing (statsmodels, scipy).

2.2 Load and Prepare Data

```
df_raw = pd.read_csv('WorldEnergy.csv')
selected_countries = ['Indonesia', 'Malaysia', 'Philippines', 'Singapore', 'Thailand',
'Vietnam']
df = df_raw[df_raw['country'].isin(selected_countries)][['country', 'year',
'low_carbon_energy_per_capita']].dropna()
df = df[df['year'] >= 2000].reset_index(drop=True)
```

- **Loads Dataset:** Reads the raw global energy data from a CSV file.
- **Filters Data:** Isolates the data for six specific Southeast Asian nations and retains only the relevant low-carbon energy columns.
- **Cleans Data:** Removes rows with missing values (`dropna()`) and restricts the timeline to the year 2000 and beyond.

2.3 Assign Categories (GDP Level and Period)

```
gdp_mapping = {'Singapore': 'High', 'Malaysia': 'Middle', 'Thailand': 'Middle',
               'Indonesia': 'Low', 'Philippines': 'Low', 'Vietnam': 'Low'}
df['gdp_level'] = df['country'].map(gdp_mapping)
df['period'] = df['year'].apply(lambda x: 'Early' if 2000 <= x <= 2010 else 'Recent')
```

- Creates GDP Variable: Maps each country to a categorical economic tier (High, Middle, or Low).
- Creates Time Variable: Divides the timeline into two distinct eras: 'Early' (2000–2010) and 'Recent' (post-2010) to allow for temporal comparisons.

2.4 One-Way ANOVA

```
groups = [group['low_carbon_energy_per_capita'].values for name, group in
df.groupby('gdp_level')]

f_stat, p_val = stats.f_oneway(*groups)
print(f"F-statistic: {f_stat:.4f}, p-value: {p_val:.4e}")
if p_val < 0.05:
    tukey = pairwise_tukeyhsd(df['low_carbon_energy_per_capita'], df['gdp_level'],
alpha=0.05)
    print(tukey)
```

- **Hypothesis Testing:** Performs a One-Way ANOVA to determine if there is a statistically significant difference in mean low-carbon energy consumption across the three GDP levels.
- **Post-Hoc Analysis:** Automatically runs a Tukey HSD test if the ANOVA is significant ($p < 0.05$) to pinpoint exactly which specific GDP groups differ from one another.

2.5 Two-Way ANOVA

```
model = ols('low_carbon_energy_per_capita ~ C(gdp_level) * C(period)', data=df).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print(anova_table)
```

- **Complex Modeling:** Fits an Ordinary Least Squares (OLS) regression model to the data.
- **Interaction Testing:** Performs a Two-Way ANOVA to test not only the individual effects of gdp_level and period on energy consumption, but also their interaction (whether the effect of time differs depending on the GDP level).

2.6 Visualizations

```
plt.figure(figsize=(14, 6))
plt.subplot(1, 2, 1)
sns.boxplot(x='gdp_level', y='low_carbon_energy_per_capita', data=df, order=['Low',
'Middle', 'High'])
plt.title('Low-Carbon Energy by GDP Level')
plt.subplot(1, 2, 2)
sns.pointplot(x='period', y='low_carbon_energy_per_capita', hue='gdp_level', data=df,
capsize=.1)
plt.title('Interaction: GDP Level and Period')
plt.show()
```

- **Distribution Boxplot**: Visually compares the spread, median, and outliers of energy consumption across the three GDP tiers.
- **Interaction Pointplot**: Illustrates the Two-Way ANOVA results by showing how the mean consumption for each GDP group shifted between the 'Early' and 'Recent' periods.

2.7 Statistical Assumption Checks

```
residuals = model.resid
shapiro_test = stats.shapiro(residuals)
print(f"Shapiro-Wilk p-value: {shapiro_test.pvalue:.4e}")
groups = [group['low_carbon_energy_per_capita'].values for name, group in
df.groupby('gdp_level')]
levene_test = stats.levene(*groups)
print(f"Levene's p-value: {levene_test.pvalue:.4e}")
```

- **Normality Check:** Uses the Shapiro-Wilk test on the model's residuals to ensure the data is normally distributed (a key requirement for ANOVA).
- **Variance Check:** Applies Levene’s Test to confirm the assumption of "homogeneity of variance," ensuring the spread of data is roughly equal across all GDP groups

3. Hypothesis for ANOVA Tests

One-Way ANOVA (Effect of GDP Level)

- **Null hypothesis (H_0)**: The mean low-carbon energy per capita is equal across all three GDP level groups (Low, Middle, High).

$$H_0: \mu_{\text{Low}} = \mu_{\text{Middle}} = \mu_{\text{High}}$$

- **Alternative hypothesis (H_a)**: At least one GDP level group has a mean low-carbon energy per capita that differs from the others.

Two-Way ANOVA (GDP Level × Period)

- **Main effect of GDP Level**
  - H_0: No difference in mean low-carbon energy per capita among GDP levels (after controlling for period and interaction).
  - H_a: At least one GDP level differs.
- **Main effect of Period**
  - H_0: The mean low-carbon energy per capita is the same for Early (2000-2010) and Recent (2011-2024) periods.
  - H_a: The mean low-carbon energy per capita differs between the two periods.
- **Interaction effect (GDP Level × Period)**
  - H_0: The effect of GDP level on low-carbon energy per capita does **not** depend on the period – i.e., differences between income groups are consistent over time.
  - H_a: The effect of GDP level **does** depend on the period, i.e., the gap between income groups changed from the Early to Recent period.

4. Results and Discussion

4.1 One Way ANOVA

```
=== One-Way ANOVA (GDP Level) ===
F-statistic: 8.5691
p-value: 3.0192e-04

Tukey HSD Post-hoc Test:
 Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====================================================
group1 group2 meandiff p-adj   lower    upper   reject
-----------------------------------------------------
  High    Low  397.396 0.0325    26.7  768.0919   True
  High Middle 682.0704 0.0002 288.888 1075.2528   True
   Low Middle 284.6744 0.0589 -8.3865  577.7353  False
-----------------------------------------------------
```

One-Way ANOVA Result

- **F-statistic:** 8.5691
- **p-value:** 0.0003 (3.0192e-04)
- Because the p-value is well below the standard 0.05 threshold, we reject the null hypothesis. This means there is a **statistically significant difference** in low-carbon energy consumption per capita across the different GDP levels. GDP level absolutely plays a role in these metrics.

2. Tukey HSD Post-hoc Test Results

Since the overall ANOVA was significant, the Tukey test compares every group against each other to see exactly where those differences lie:

- **High vs. Low GDP:**
  - **p-value:** 0.0325
  - **Result (Reject = True):** There is a **statistically significant difference** between the High GDP and Low GDP groups. The mean difference is 397.4 kWh per capita.
- **High vs. Middle GDP:**
  - **p-value:** 0.0002
  - **Result (Reject = True):** There is a highly **statistically significant difference** between the High GDP and Middle GDP groups. The mean difference is 682.1 kWh per capita.
- **Low vs. Middle GDP:**
  - **p-value:** 0.0589
  - **Result (Reject = False):** Because the p-value is slightly above 0.05, there is **no statistically significant difference** between the Low and Middle GDP groups.

Conclusion

The **High GDP group** (Singapore) is the outlier, its low-carbon energy consumption per capita is significantly different from the rest of the region. Meanwhile, countries in the **Low** and **Middle GDP groups** have energy consumption profiles that are statistically similar to one another.

4.2 Two Way ANOVA

```
=== Two-Way ANOVA (GDP Level * Period) ===
                              sum_sq     df           F        PR(>F)
C(gdp_level)            7.876866e+06    2.0  11.984706  1.532940e-05
C(period)               1.732072e+07    1.0  52.707186  2.224542e-11
C(gdp_level):C(period)  2.919909e+06    2.0   4.442662  1.342029e-02
Residual                4.732151e+07  144.0        NaN           NaN
```

1. Main Effect: GDP Level (C(gdp_level))

- **p-value:** 0.000015 (1.53e-05)
- **Interpretation:** Because the p-value is significantly less than 0.05, there is a **statistically significant main effect** of GDP level. This means that, ignoring the time period, a country's GDP level has a significant impact on its low-carbon energy consumption.

2. Main Effect: Time Period (C(period))

- **p-value:** < 0.000001 (2.22e-11)
- **Interpretation:** Because the p-value is practically zero (well below 0.05), there is a **highly significant main effect** of the time period. This indicates that, regardless of a country's GDP, low-carbon energy consumption has changed significantly between the 'Early' (2000-2010) and 'Recent' (post-2010) periods.

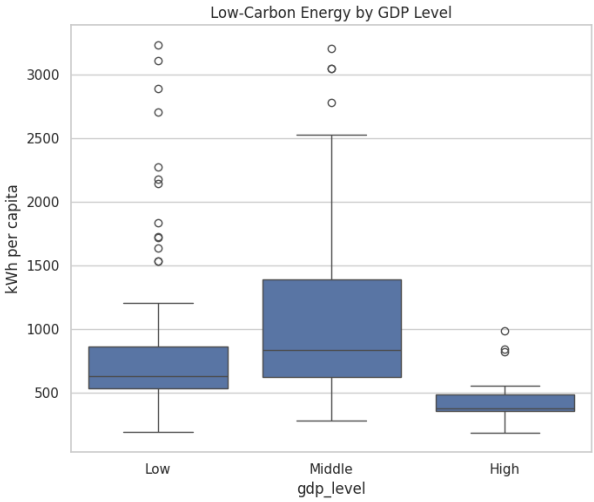
3. Interaction Effect (C(gdp_level):C(period))

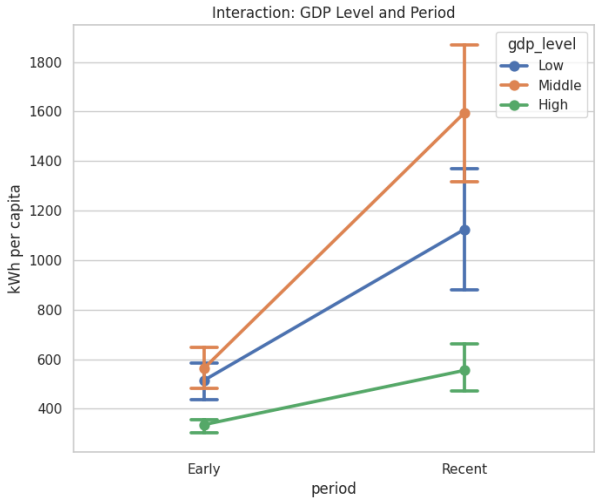
- **p-value:** 0.0134 (1.34e-02)
- **Interpretation:** Because the p-value is less than 0.05, there is a **statistically significant interaction** between GDP level and time period.

Conclusion

Not only do GDP level and period individually affect low-carbon energy consumption, but their interaction is also significant. This means the way energy consumption changed over time was not the same for all economic tiers. The jump or shift in low-carbon energy adoption between the 'Early' and 'Recent' periods depended specifically on whether a country was in the High, Middle, or Low GDP group.

4.3 Visualizations





Key Observations:

Box Plot: Low-Carbon Energy by GDP Level

This chart visualizes the distribution of energy consumption across the three economic tiers

- **Middle GDP Leads:** The 'Middle' GDP group clearly has the highest median low-carbon energy consumption per capita and the widest spread (Interquartile Range), indicating high but highly variable adoption.
- **High GDP Lags:** The 'High' GDP group has the lowest median and the tightest clustering of data points. This visually confirms the Tukey HSD result where the 'High' group was significantly lower than the others.
- **Presence of Outliers:** Both the 'Low' and 'Middle' groups have several high outliers (the empty circles above the whiskers), indicating specific years or countries within those tiers that achieved unusually high low-carbon energy consumption.

2. Interaction Plot: GDP Level and Period

This chart is the visual representation of your Two-Way ANOVA results, specifically highlighting how the groups changed over time.

- **Universal Growth (Main Effect of Period):** All three lines slope upward from the 'Early' to the 'Recent' period. This visualizes the highly significant main effect of time, every group increased their low-carbon energy consumption per capita.
- **Non-Parallel Lines (The Interaction Effect):** The most important feature of an interaction plot is whether the lines are parallel. Because these lines are clearly diverging (spreading apart), it visually confirms the significant interaction p-value ($p = 0.0134$). The time period affected the groups differently.
- **Diverging Growth Rates:** The **Middle GDP** group experienced the sharpest, most aggressive acceleration in low-carbon energy adoption between the two periods (the steepest slope).

- The **Low GDP** group also showed strong growth, but slightly less steep than the Middle group.
- The **High GDP** group experienced the weakest growth (the flattest slope), falling further behind the other two groups in the 'Recent' period compared to where they started in the 'Early' period.

4.4 Assumption Checks

4.4.1 Shapiro-Wilk Test

```
=== Shapiro-Wilk Test (Residuals) ===
Statistic: 0.8961
p-value: 7.9662e-09
Result: Residuals do not appear to be normally distributed (reject H0).
```

Shapiro-Wilk Test (Normality of Residuals)

- Purpose: This test checks a fundamental assumption of ANOVA: that the residuals (the differences between the observed values and the model's predicted values) follow a normal (bell-shaped) distribution. The null hypothesis (H_o) is that the residuals *are* normally distributed.
- p-value: 7.9662e-09 (which is approximately 0.000000008)
- Interpretation: Because the p-value is extremely small, far below the standard 0.05 threshold, we reject the null hypothesis.

4.4.2 Levene's Test

```
=== Levene's Test (Homogeneity of Variance) ===
Statistic: 5.3693
p-value: 5.6156e-03
Result: Variances are not equal across groups (reject H0).
```

Levene's Test (Homogeneity of Variance)

- Purpose: This test checks another critical assumption of ANOVA: that the variance (the spread or dispersion of the data) is roughly equal across all the different groups you are comparing (in this case, the GDP levels). The null hypothesis (H0) is that the variances are equal across all groups.
- Test Statistic & p-value: The test yielded a statistic of 5.3693 and a p-value of 5.6156e-03 (which is approximately 0.0056).
- Interpretation: Because the p-value is significantly less than the standard 0.05 significance level, we reject the null hypothesis.